

AI Data Center Committee Power Generation Scale and Siting June 10th, 2026 Public Hearing

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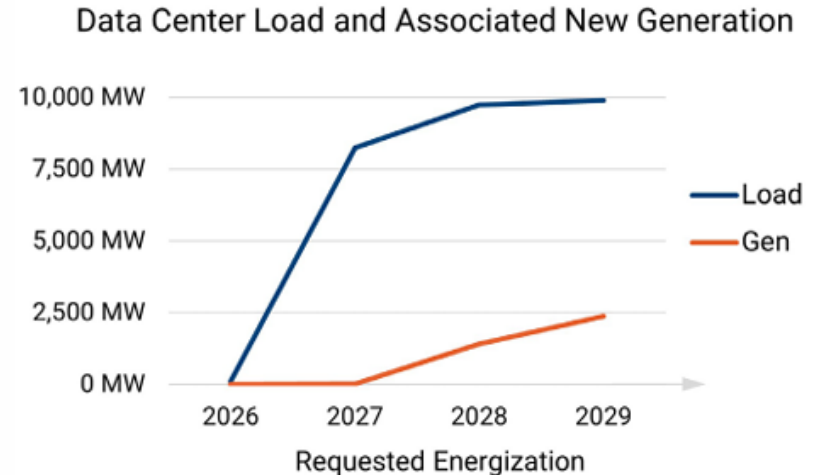
AI Visualization of a 35 ft tall Data Center and Natural Gas Power Generating Facility

Power Requirement

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Due to the high demand for power for AI datacenters in Alberta the AESO has effectively “capped” the connection availability to 1200MW until 2028 and have already selected the 2 projects that will use this allotment. Projects outside of this will be required to BYOP. (Bring Your Own Power)

This will require that the project proposed for Vulcan County will need to generate a good portion of its own electricity which will require a significant power generating facility be installed for the build out of the complex. Once the location is selected it will need to meet the future 2000MW requirements.



[Source: AESO Data Centre Update](#)

Scale of Power Required

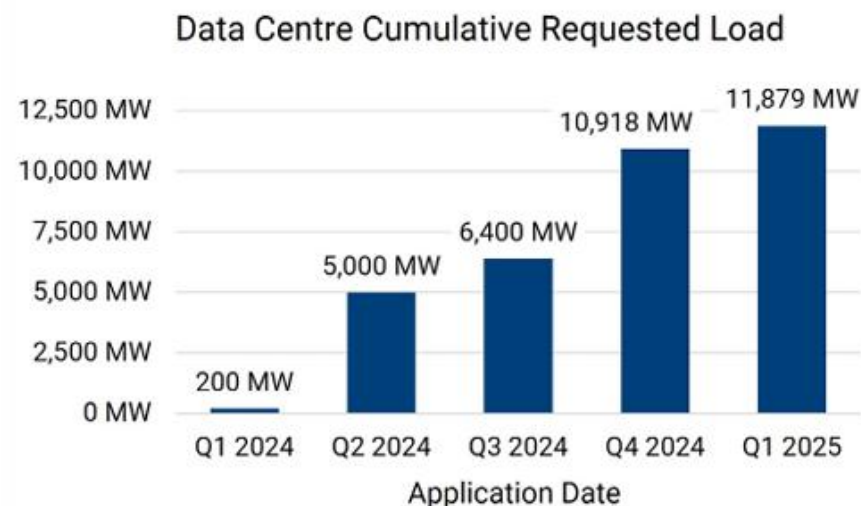
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The proposed facility has proposed the first phase requiring 500MW with an ultimate build out to 2000MW (2GW)

For Context:

City of Edmonton approximate load:
1400MW

City of Calgary: 1824MW highest peak
load (July 23, 2024)



Sources:

[AESO Data Centre Update](#)

[Corporate Energy Consumption | Open Calgary](#)

[AESO Announces Interim Approach to Large Load Connections » AESO](#)

Scale of Power Required

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Some Examples of Current Power
Generating Facilities in Alberta:

Shepard: 869MW

Combined Cycle Natural Gas Turbines
and Steam Turbine

Sources:

ets.aeso.ca/ets_web/ip/Market/Reports/CSDReportServlet

[The Shepard Energy Centre | Electricity Canada](#)



Scale of Power Required

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Some Examples of Current Power
Generating Facilities in Alberta:

Around the 500MW Range:

Genesse Repower 1: 466MW

Genesse Repower 2: 466MW

Natural Gas Turbines with Heat
Recovery and Steam Turbines



Source:

<https://www.majorprojects.alberta.ca/details/Genesse-Power-Plant-Natural-Gas-Conversion/4403>

Scale of Power Required

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Some Examples of Current Power Generating Facilities in Alberta:

Solar:

Travers: 465MW*

Wind:

Buffalo Plains Wind Farm: 495MW*

*Ultimate capacity in ideal conditions, not continuous and obviously not at night related to solar



Source:

<https://majorprojects.alberta.ca/details/Travers-Solar-Project/3656>

Power Options - Continuous

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Large natural gas turbines, generally with heat recovery systems and steam turbine.



Smaller turbines (only 1MW each) –likely requiring 100s of units

Sources:

[Genesee Energy Centre: the heart of Alberta's decarbonized future | Electricity Canada](#)

[Products :: Capstone Power Solutions](#)



Expandable

Can be paralleled up to 30MW of power

Emergency Power Scale

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Application lists full facility load required with 99.99999% uptime.

Diesel Gensets: 1-4MW (each)

Context: Engines used for Trains

Large seacan size x 124 units (at least)

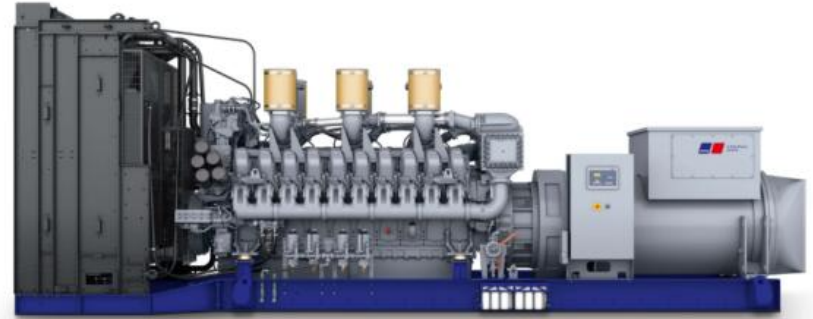
5 Million Litres of Diesel (listed on application)

Tank shown from Avenell project, note circled area that shows a worker located by the valve for scale of size.

Sources:

[Diesel Generator Sets](#)

[Project: Avenell – 5 Million Litre Diesel Tank - Sun Engineering](#)



New Info In Context

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1,500,000 MWh annual curtailment is equivalent to 171MW assuming that the generating and batteries were being filled in an optimal manner and dependent on weather conditions year to year.

Power Plant true size should be clear prior to siting being completed, website notes a potential range of 100MW (substantial) on Phase 1 alone which then has the remaining project phases bolted on top. The remaining phases are potentially not local renewable resources given the current position of the Alberta government on renewables on agricultural land.

Other power from renewables in the area already in use by province and therefore shortfall will need to be generated within the project.

Hydrogen fuel cells, storage and potential compression added to project scope

Palliser Grid		
Opportunity Site Phases Technology Community Founder Invest		
We convert curtailed renewable energy into AI compute and pay the demand tariffs that fund congestion relief for the region.		
LINE	CONGESTED HOURS	ANNUAL CURTAILMENT
CBW Path (Cassils-Bowmanton-Whitla)	61%	1,380,000 MWh
924L (Milo)	10%	41,000 MWh
927L (Milo)	9%	37,000 MWh
935L (Milo)	2%	6,000 MWh
9L59 (Anderson)	11%	36,000 MWh
Southern Alberta total		~1,500,000 MWh

Source: [Palliser Grid Website](#)

Inappropriately Sited

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Consideration required that this is an industrial power plant and not merely warehouses that contain servers, cooling systems and pumps. Is there a more appropriate location within the county for this and has enough pre-work been done to know for certain that this is the optimal location?

The original designation related to datacenters and even bitcoin mining did not contemplate the level of power usage that an AI datacenter presents.

